

1. Introduction

“Unit testing isn’t a new concept in software development. It’s been floating around since the early days of the Smalltalk programming language in the 1970s, and it proves itself time and time again as one of the best ways a developer can improve code quality while gaining a deeper understanding of the functional requirements of a class or method.” [Osherove 2013]

“Kent Beck introduced the concept of unit testing in Smalltalk, and it has carried on into many other programming languages, making unit testing an extremely useful practice in software programming.” [Osherove 2013]

Roy Osherove gives a practical definition of “unit test” in [Osherove 2013] as follows:

“A unit test is an automated piece of code that invokes the unit of work being tested, and then checks some assumptions about a single end result of that unit. A unit test is almost always written using a unit testing framework. It can be written easily and runs quickly. It’s trustworthy, readable, and maintainable. It’s consistent in its results as long as production code hasn’t changed.”

2. Learning Objectives

- creating and running unit tests using Microsoft Visual Studio IDE
- viewing code coverage using Microsoft Visual Studio IDE

3. Prerequisites

You should have gone over the chapter 1 of [Osherove 2013] before this lab session.

4. Instructions

1. Please read the document titled “Sample Project for Creating Unit Tests” [1]. Please explore and understand the sample code (Note: The only intentional error in the sample code is that the in Debit method "m_balance += amount" should have a minus not a plus sign before the equals sign).
2. Please read the document titled “Walkthrough: Creating and Running Unit Tests for Managed Code” [2] and go through the steps to create and customize unit tests, run them, and examine the test results.
3. Please read the document titled “Walkthrough: Run tests and view code coverage” [3] and view code coverage data using Microsoft Visual Studio IDE.

5. References

[Osherove 2013] The Art of Unit Testing with examples in C#, 2nd Ed., Roy Osherove, Manning Publications, 2013.

[1] Sample Project for Creating Unit Tests, <http://msdn.microsoft.com/en-us/library/ms243176.aspx>.

[2] Walkthrough: Creating and Running Unit Tests for Managed Code, <http://msdn.microsoft.com/en-us/library/ms182532.aspx>.

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Lab 4

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[3] Walkthrough: Run tests and view code coverage, <http://msdn.microsoft.com/en-us/library/ms182534.aspx>.